CS23431 - OPERATING SYSTEM

EXP 9: Best Fit Contiguous Memory Allocation Using C

QUESTION :

Write a C program to implement the Best Fit Contiguous Memory Allocation technique.The program should accept the sizes of memory blocks and the sizes of processes as input.

For each process, the program searches all available memory blocks and allocates the smallest block that is sufficient to satisfy the process size.If no suitable block is available, the process is marked as not allocated.The program should display the allocation details for each process and calculate the internal fragmentation.

CODE:

```
#include <stdio.h>

int main() {

   int m, n;

   printf("Enter number of memory blocks:\n");
   scanf("%d", &m);

 =

   int blockSize[m];    // stores remaining size of each block
   int original[m];     // stores original size for display



   for(int i = 0; i < m; i++) {
       scanf("%d", &blockSize[i]);



       original[i] = blockSize[i];
   }
```

```
    printf("Enter number of processes:\n");
    scanf("%d", &n);


    int processSize[n];


    printf("Enter size of each process:\n");

    for(int i = 0; i < n; i++) {
        scanf("%d", &processSize[i]);
    }

    for(int i = 0; i < n; i++) {

        int bestIndex = -1;


        for(int j = 0; j < m; j++) {

            if(blockSize[j] >= processSize[i]) {


                if(bestIndex == -1 || blockSize[j] <
blockSize[bestIndex]) {
                    bestIndex = j;
                }
            }
        }

}
```

Output:

```
5
100 500 200 300 600
4
212 417 112 426
```

```
Enter number of memory blocks:
Enter size of each block:
Enter number of processes:
Enter size of each process:
Process 1 of size 212 is allocated to Block 4 of size 300
with Fragment 88
Process 2 of size 417 is allocated to Block 2 of size 500
with Fragment 83
Process 3 of size 112 is allocated to Block 3 of size 200
with Fragment 88
Process 4 of size 426 is allocated to Block 5 of size 600
with Fragment 174
```

Output :

```
Enter number of memory blocks:
Enter size of each block:
Enter number of processes:
Enter size of each process:
Process 1 of size 212 is allocated to Block 4 of size 300 with Fragment 88
Process 2 of size 417 is allocated to Block 2 of size 500 with Fragment 83
Process 3 of size 112 is allocated to Block 3 of size 200 with Fragment 88
Process 4 of size 426 is allocated to Block 5 of size 600 with Fragment 174
```